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Multimodal AI for Body Fat Estimation: Computer Vision and Anthropometry with DEXA Benchmarks

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Abstract

AI-based models were tested as low-cost alternatives to DEXA scans for body fat estimation. Using 535 samples (253 anthropometric, 282 image), two models were trained:

- CNN (ResNet)** on body images → $R^2 = 0.81$, RMSE = 4.44%, MAE = 3.34%
- Regression** on body measurements → $R^2 = 0.57$, RMSE = 4.47%, MAE = 3.52%

CNN significantly outperformed traditional methods (BMI, Navy), showing potential for accessible health tracking.

Introduction

Importance of Body Fat %1

- Key indicator** of overall health, fitness, and disease risk.
- Problem:** DEXA scans are accurate but expensive and inaccessible, while BMI and Navy methods lack precision.
- Opportunity:** Advances in AI and computer vision can bridge the gap between accuracy and accessibility.
- Goal:** Evaluate whether AI models can estimate BF% near DEXA-level accuracy while remaining affordable and scalable to the public.

Correlation with Body Fat (%)

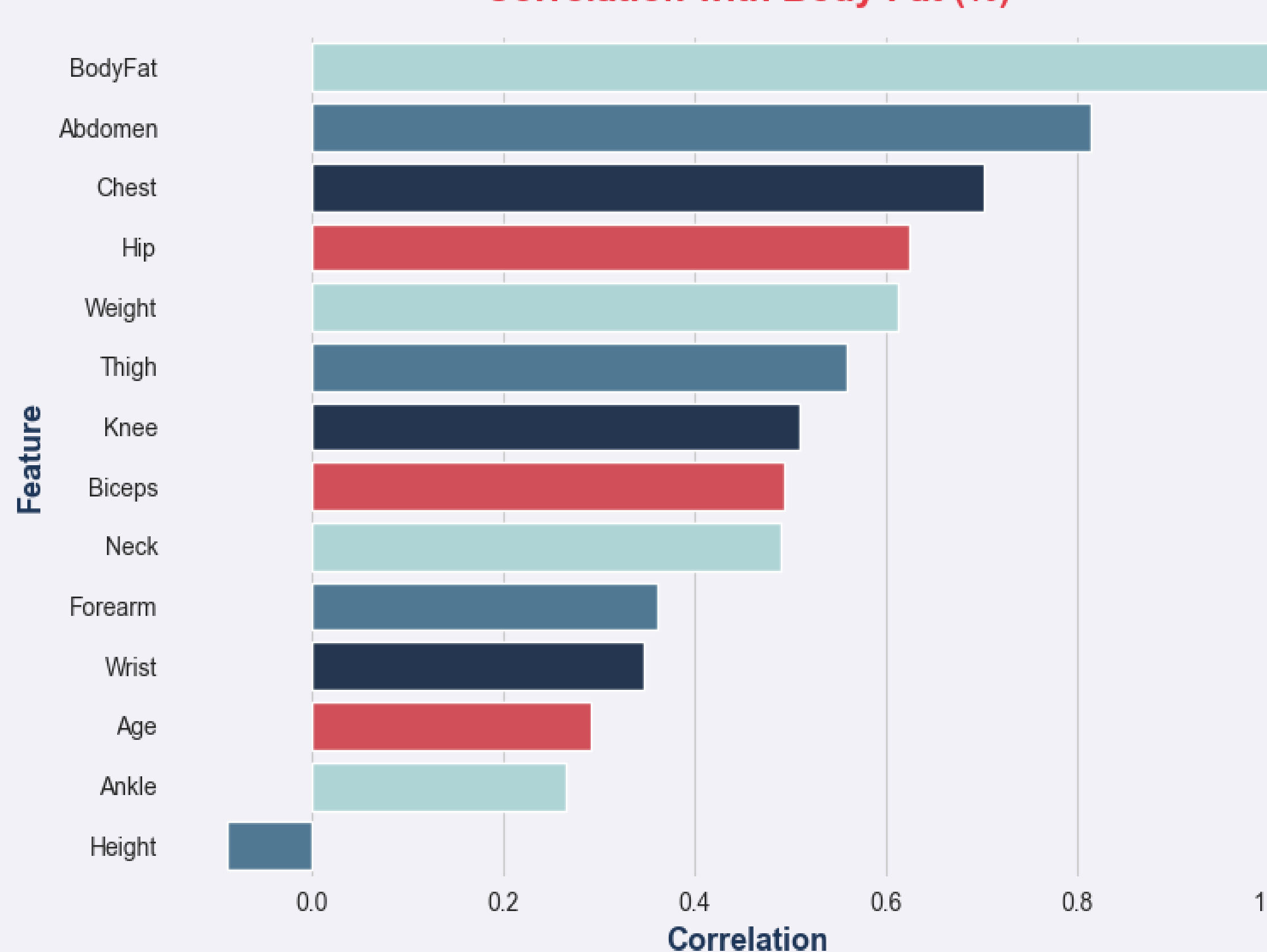


Figure 1 shows the correlation between Bodyfat and different body measurements

Methodology

Regression using anthropometric data:

- Inputs:** body measurements (weight, chest, abdomen, hip, thigh)
- Loss:** Mean Squared Error (MSE)
- Metrics:** MAE, RMSE, R^2

CNN using image data

- Architecture:** Convolutional Neural Network based on ResNet
- Inputs:** frontal body images

Goal:

- Extract** visual features describing body composition
- Evaluation**
- Both models were evaluated on holdout sets
- Both models were not only compared to each other, but also against baseline methods (**BMI & Navy equation**)

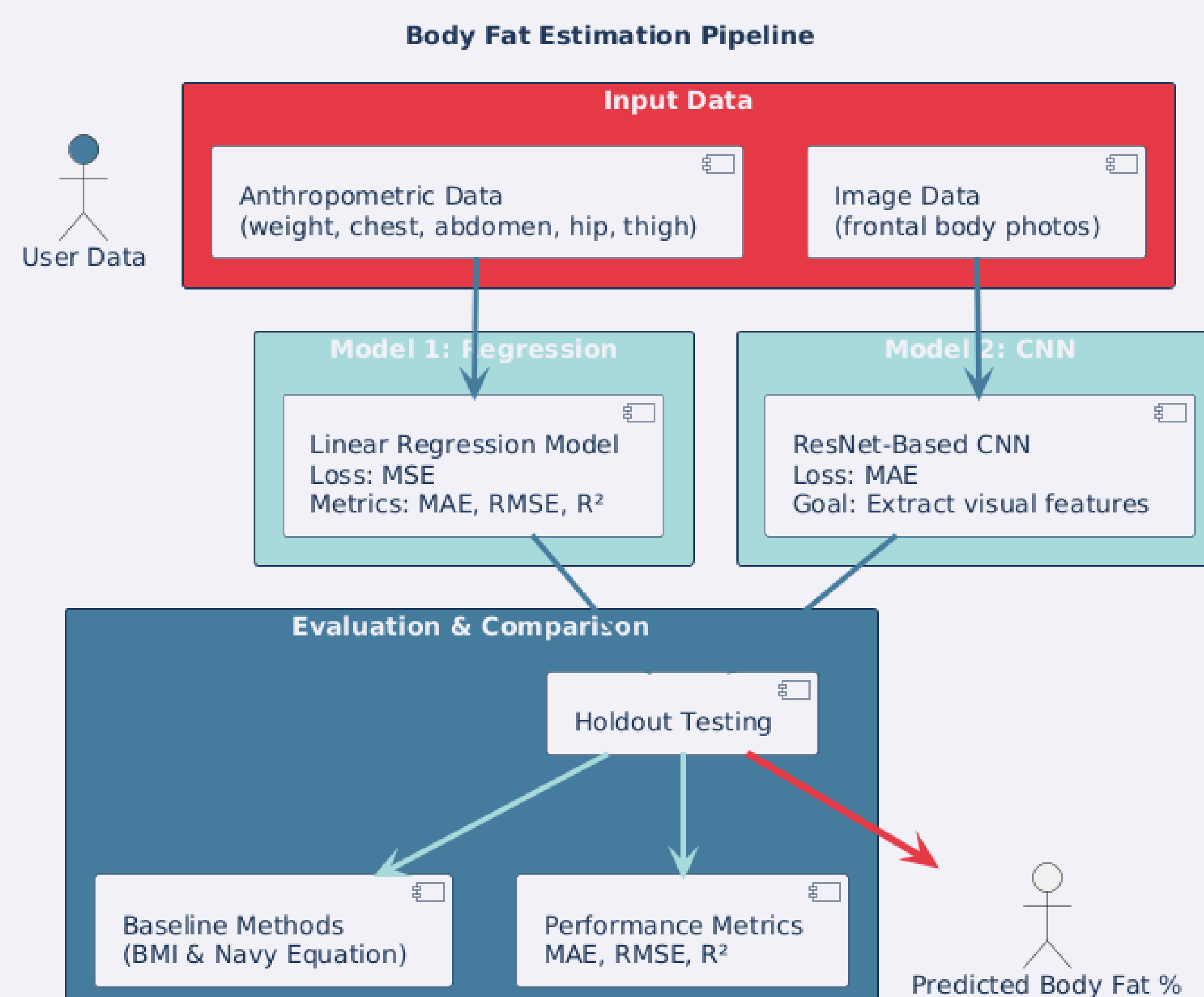


Figure 2 shows the overall body fat estimation pipeline, outlining both regression and CNN models from data input to evaluation and prediction.

Dataset

Description	Anthropometric Data n = 253	Image Data n = 282
Body Fat (%)	19.1 ± 8.3	20.0 ± 10.3
Height (cm)	178.2 ± 3.7	N/A
Weight (kg)	80.7 ± 29.3	N/A
Data Type	Physical measurements	Frontal body photographs
Model Applied	Linear Regression (MSE loss)	CNN (ResNet-based, MAE loss)

References

- S. B. Heymsfield et al., *Human Body Composition: Advances in Models and Methods*, Annu. Rev. Nutr., vol. 17, pp. 527–558, 1997.
- J. A. Hodgdon and M. B. Beckett, *Prediction of Percent Body Fat for U.S. Navy Men from Body Circumferences and Height*, Naval Health Research Center Report No. 84-11, 1984.

Results

Anthropometric Regression Model

- Accuracy:** $R^2 = 0.57$, RMSE = 4.47%
- Strongest predictor:** abdominal circumference
- Moderate performance, limited at extreme BF% ranges

Image-Based CNN Model

- Accuracy:** $R^2 = 0.81$, RMSE = 4.44%
- Recognized visual patterns** in addition to the measurements
- Scatter plots suggested **good mid-range** predictions, but variability at extremes
- Loss curves:** stable convergence, with mild overfitting (small dataset)

Comparison

- Both AI models **outperformed** heuristic baselines (BMI, Navy formula)
- CNN** provided the best estimates overall

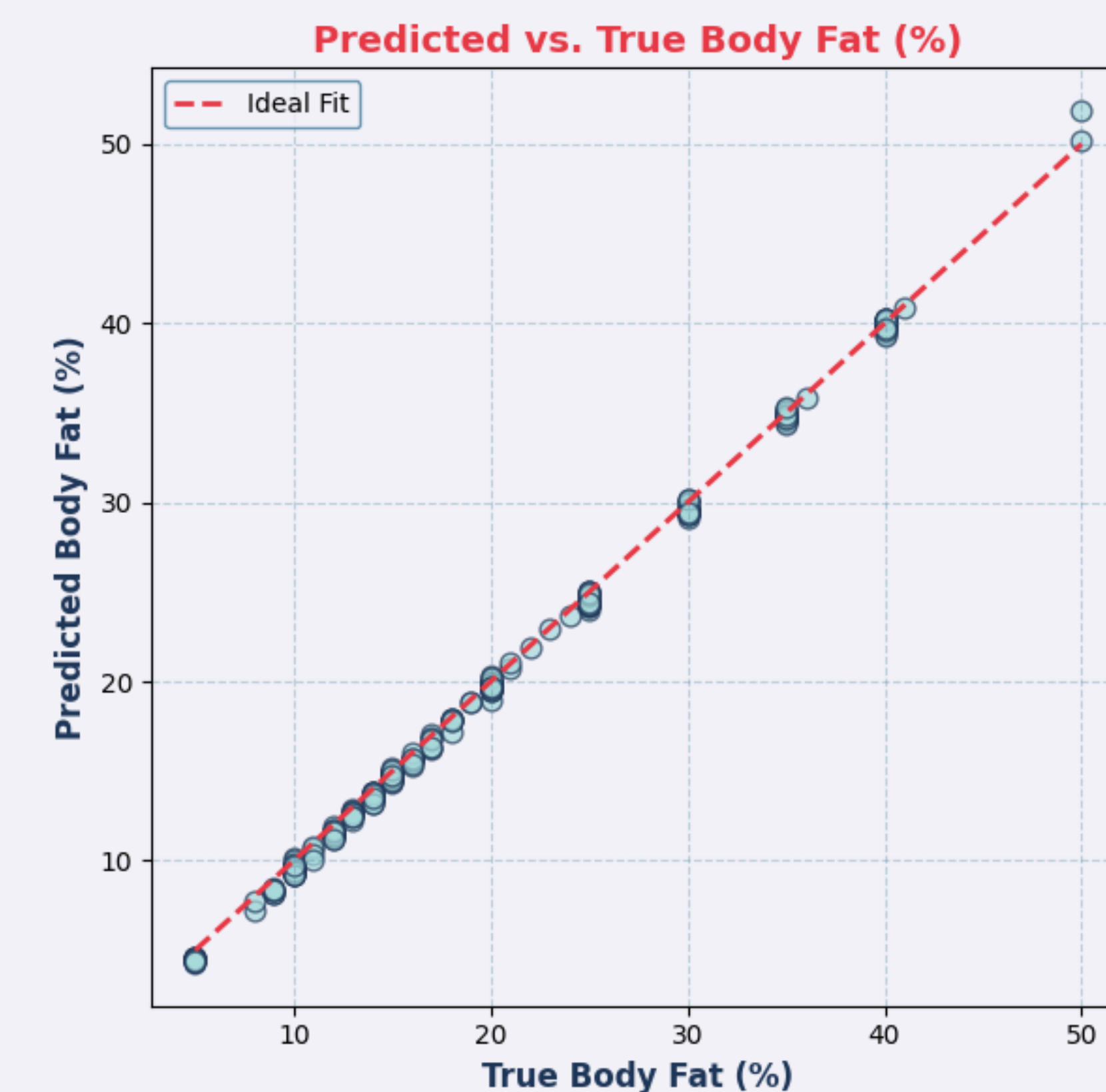


Figure 3 shows a strong linear relationship between predicted and true body fat values, indicating high model accuracy and consistency.

Model Performance Comparison

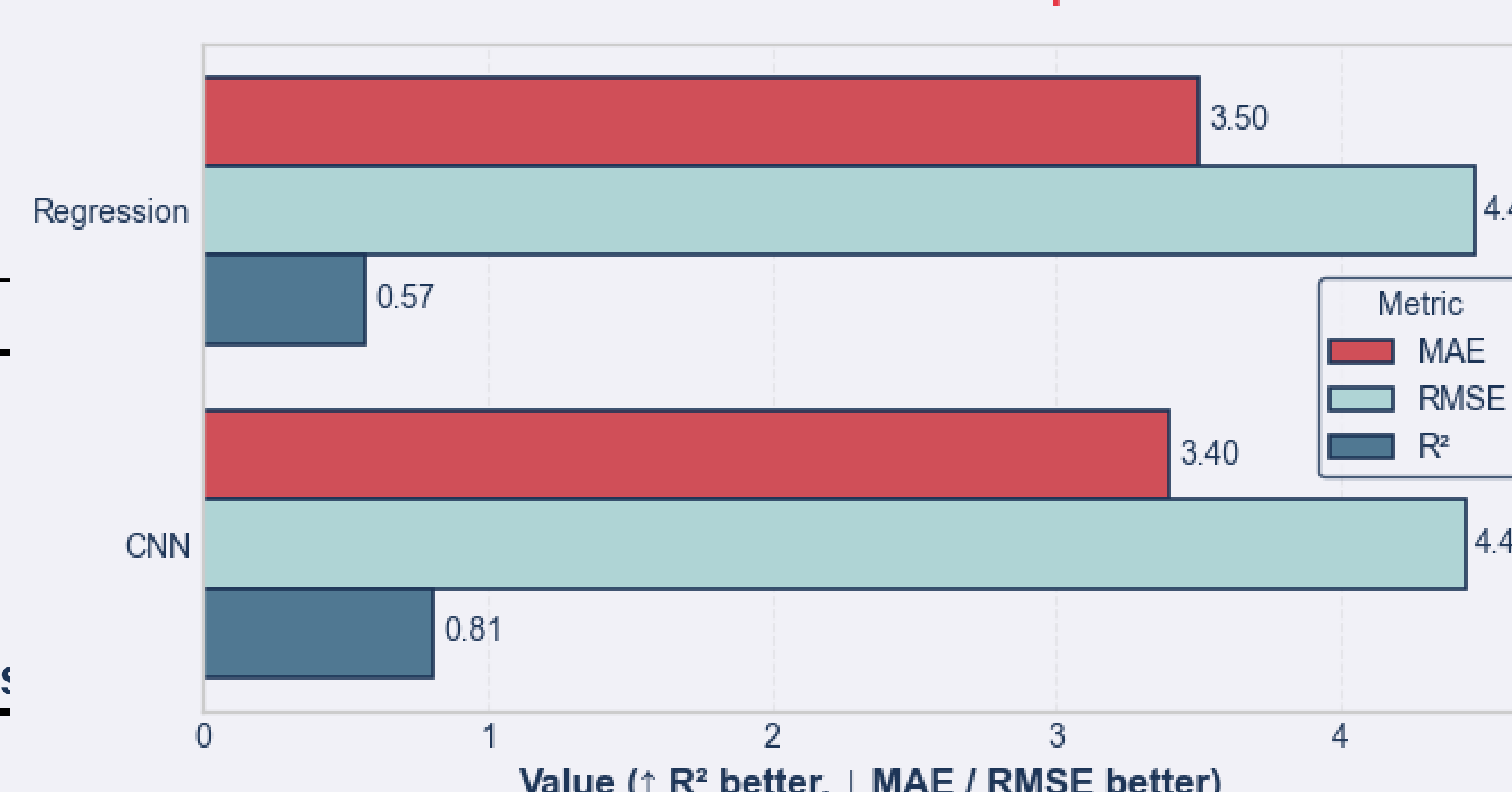


Figure 4 shows the different results after training both models, highlighting the superiority of the Image-Based Model over Anthropometric Model

Conclusion

- Feasibility:** AI approaches have a good prospect for inexpensive body fat estimation.
- Performance:** The CNN model ($R^2 = 0.81$) outperformed the regression model ($R^2 = 0.57$) and baseline methods.
- Weaknesses:** Small sample size, only anthropometric records from males, and reliance on self-reported BF% values.
- Substance:** This represents a meaningful step towards the development of scalable, AI-supported health tools to help create equity in tracking of body composition. However, validating against large multimodal datasets will be an important next step.

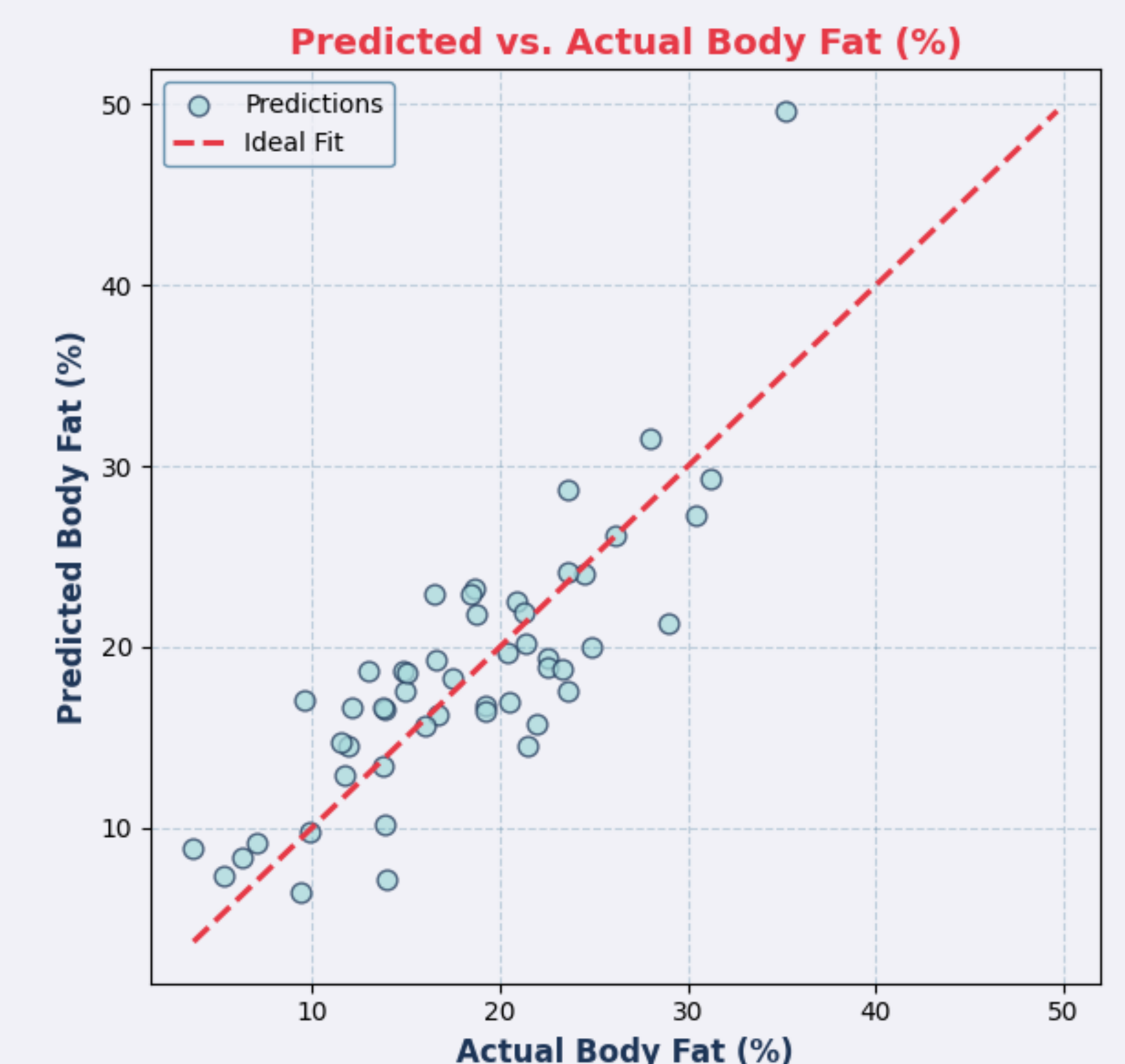


Figure 5 shows a positive correlation between predicted and actual body fat values, though with greater variability across samples.

Future Work

This research indicated that AI-based methods for estimating body fat can be developed, however, the following avenues remain for future research:

- Paired Multimodal Data** - Data sets that include both image data and anthropometric measurements taken from the same subjects so true multimodal fusion models can be created.
- Larger and More Diverse Sample Cohort** - More data will be available if samples include females, variability in ethnicities, and diverse body types, improving the generalizability of the methods in this thesis.
- Academic Validation** - Future work should include comparison of our estimates against gold standard measures (DEXA, MRI) through controlled study designs to establish the academic validity.